

Oscar J. Escobar

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SKILLS

- Skills
 - Deep Learning & Machine Learning
 - Natural Language Processing
 - Deep Reinforcement Learning
 - Optimal Control & Optimization
 - Fourier Analysis & Computational Linear Algebra
 - Bayesian Statistics & Probability Theory
 - Time Series Analysis & Hidden Markov Models
 - Data Assimilation, PDEs, ODEs
- Coding/Software:
 - Expert: Python, Git/GitHub, Unix Shell/Bash
 - Proficient: SciKit Learn, PyTorch & Keras, Gymnasium, CVXPY, Pandas, SQL
 - Experience with: PySpark, RLLib, Oracle, RStudio, Hadoop, Excel & Power BI, C++

PROFESSIONAL EXPERIENCE

Machine Learning Intern Developer

June 2024 — Aug 2024

Wells Fargo—Advanced Analytics & Solutions Team

Minneapolis, MN

- Crafted NLP text classification model that achieved 77% cross-validated accuracy on high-error-prone data
- Leveraged NLP and statistical analysis to perform feature engineering and presented NLP results and findings to team
- Wrote an executive summary detailing overall team results, findings, and suggestions presented to upper management

Machine Learning & Data Science Student Engineer

April 2023 — April 2024

Family Search—Automated Content Extraction Team

Lehi, UT

- Performed data analysis on ML model output using self-made Python script that compared over 100 records at a time
- Identified model improvement areas that led to an increase in correct predictions of 20-30% using Python script
- Spearheaded efforts and meetings with a team of 3 data labelers to create training data and define project benchmarks

RESEARCH

Research Assistant

Sept 2024 — Present

BYU Mathematics Department

Provo, UT

- Optimize wildfire modeling equations using the Fourier Basis and computational linear algebra
- Craft techniques to employ the Fast Fourier Transform algorithm

Subteam Lead & Analyst

Sept 2021 — Aug 2023

BYU Rocketry Club—Hybrid Rocket Research Team

Provo, UT

- Published critical design review and research in [Digital Commons Utah State University](#)
- Crafted a Python script modeling 3 different axisymmetric nozzles incorporating Computational Fluid Dynamics

PROJECTS

Breast Cancer Modeling & Optimal Dosage

Sept 2024 — April 2025

- Modeled Invasive Ductal Carcinoma treatment that met the American Cancer Society timelines of chemotherapy
- Applied optimal control theory to derive continuous AC treatment dosage

Deep Reinforcement Learning

Dec 2025 — Present

- Implemented code and theory of DQN, DDQN, and Dueling Architecture for Atari game play using CNNs and images
- Developing and coding VPG and Open AI's PPO mathematical theory and algorithm for Humanoid Robotic Walker

EDUCATION

Bachelor of Science: Applied and Computational Mathematics

Dec 2025

Brigham Young University (Major GPA 3.3)

Provo, UT

- 2023-2024 Math Class President; BYU SIAM Chapter Vice-president; AI & Data-Science Club Member
- Minor/Concentration: Machine Learning
- Related coursework:
 - Mathematics of Deep Learning
 - Data Assimilation
 - Modeling with Dynamics & Data
 - NLP